



help binsreg

Title

binsreg — Data-Driven Binscatter Least Squares Estimation with Robust Inference Procedures and Plots.

Syntax

```
binsreg depvar indvar [othercovs] [if] [in] [weight] [ , deriv(v) at(position)
      absorb(absvars) reghdfeopt(reghdfe_option)
      dots(dotsopt) dotsgrid(dotsgridoption) dotsplotopt(dotsoption)
      line(lineopt) linegrid(#) lineplotopt(lineoption)
      ci(ciopt) cigrid(cigridoption) ciplotopt(rcapoption)
      cb(cbopt) cbgrid(#) cbplotopt(rareaoption)
      polyreg(p) polyreggrid(#) polyregcigrid(#) polyregplotopt(lineoption)
      by(varname) bycolors(colorstylelist) bysymbols(symbolstylelist)
      bylpatterns(linepatternstylelist)
      nbins(nbinsopt) binspos(position) binsmethod(method) nbinsrot(#)
      samebinsby randcut(#)
      pselect(numlist) sselect(numlist)
      nsims(#) simsgrid(#) simsseed(seed)
      dfcheck(n1 n2) masspoints(masspointsoption)
      vce(vcetype) asyvar(on/off)
      level(level) usegtools(on/off) precision(single/double) noplot
      savedata(filename) replace
      plotxrange(min max) plotyrange(min max) twoway_options ]
```

where *depvar* is the dependent variable, *indvar* is the independent variable for binning, and *othercovs* are other covariates to be controlled for.

The degree of the piecewise polynomial *p*, the number of smoothness constraints *s*, and the derivative order *v* are integers satisfying $0 \leq s, v \leq p$, which can take different values in each case.

fweights, **awweights** and **pweights** are allowed; see [weight](#).

Description

binsreg implements binscatter least squares estimation with robust inference procedure and plots, following the results in [Cattaneo, Crump, Farrell and Feng \(2024\)](#) and [Cattaneo, Crump, Farrell and Feng \(2026\)](#). Binscatter provides a flexible way to describe the mean relationship between two variables, after possibly adjusting for other covariates, based on partitioning/binning of the independent variable of interest. The main purpose of this command is to generate binned scatter plots with curve estimation with robust pointwise confidence intervals and uniform confidence band. If the binning scheme is not set by the user, the companion command [binsregselect](#) is used to implement binscatter in a data-driven (optimal) way. Hypothesis testing for parametric specifications of and shape restrictions on the regression function can be conducted via the companion command [binstest](#). Hypothesis testing for pairwise group comparisons can be conducted via the companion command [binspwc](#).

A detailed introduction to this command is given in [Cattaneo, Crump, Farrell and Feng \(2025\)](#). Companion R and Python packages with the same capabilities are available (see website below).

Companion commands: [binstest](#) for hypothesis testing for parametric specifications and shape restrictions, [binspwc](#) for hypothesis testing for pairwise group comparisons, [binsglm](#) for generalized linear model binscatter estimation and inference, and [binsregselect](#) for data-driven binning selection.

Related Stata, R and Python packages are available in the following website:

<https://nppackages.github.io/>

Options

Estimand

deriv(v) specifies the derivative order of the regression function for estimation and plotting. The default is **deriv(0)**, which corresponds to the function itself.

at(position) specifies the values of *othercovs* at which the estimated function is evaluated for plotting. The default is **at(mean)**, which corresponds to the mean of *othercovs*. Other options are: **at(median)** for the median of *othercovs*, **at(0)** for zeros, and **at(filename)** for particular values of *othercovs* saved in another file.

Note: When **at(mean)** or **at(median)** is specified, all factor variables in *othercovs* (if specified) are excluded from the evaluation (set as zero).

Reghdfe

absorb(absvars) specifies categorical variables (or interactions) representing the fixed effects to be absorbed. This is equivalent to including an indicator/dummy variable for each category of each *absvar*. When **absorb()** is specified, the community-contributed command **reghdfe** instead of the command **regress** is used.

reghdfeopt(reghdfe_option) options to be passed on to the command **reghdfe**.

Important:

1. Fixed effects added via **absorb()** are included in the estimation procedure but excluded from the evaluation of the estimated function (set as zero), regardless of the option specified within **at()**. To plot the binscatter function for a particular category of interest, save the values of *othercovs* at which the function is evaluated in another file, say, **wval.dta**, specify the corresponding factor variables in *othercovs* directly, and add the option **at(wval)**.
2. **absorb()** and **vce()** should not be specified within **reghdfeopt()**.
3. Make sure the package **reghdfe** installed has a version number greater than or equal to 5.9.0 (03jun2020). An older version may result in an error in Mata.

For more information about the community-contributed command **reghdfe**, please see <https://scorreia.com/software/reghdfe/>.

Dots

dots(dotsopt) sets the degree of polynomial and the number of smoothness for point estimation and plotting as "dots". If **dots(p s)** is specified, a piecewise polynomial of degree *p* with *s* smoothness constraints is used. The default is **dots(0 0)**, which corresponds to piecewise constant (canonical binscatter). If **dots(T)** is specified, the default **dots(0 0)** is used unless the degree *p* or smoothness *s* selection is requested via the option **pselect()** or **sselect()** (see more details in the explanation of **pselect()** and **sselect()**). If **dots(F)** is specified, the dots are not included in the plot.

dotsgrid(dotsgridoption) specifies the number and location of dots within each bin to be plotted. Two options are available: *mean* and a *numeric* non-negative integer. The option **dotsgrid(mean)** adds the sample average of *indvar* within each bin to the grid of evaluation points. The option **dotsgrid(#)** adds # number of evenly-spaced points to the grid of evaluation points for each bin. Both options can be used simultaneously: for example, **dotsgrid(mean 5)** generates six evaluation points within each bin containing the sample mean of *indvar* within each bin and five evenly-spaced points. Given this choice, the dots are point estimates evaluated over the selected grid within each bin. The default is **dotsgrid(mean)**, which corresponds to one dot per bin evaluated at the sample average of *indvar* within each bin (canonical binscatter).

dotsplotopt(dotsoption) standard graphs options to be passed on to the **twoway** command to modify the appearance of the plotted dots.

Line

line(lineopt) sets the degree of polynomial and the number of smoothness constraints for plotting as a "line". If **line(p s)** is specified, a piecewise polynomial of degree p with s smoothness constraints is used. If **line(T)** is specified, **line(0 0)** is used unless the degree p or smoothness s selection is requested via the option **pselect()** or **sselect()** (see more details in the explanation of **pselect()** and **sselect()**). If **line(F)** or **line()** is specified, the line is not included in the plot. The default is **line()**.

linegrid(#) specifies the number of evaluation points of an evenly-spaced grid within each bin used for evaluation of the point estimate set by the **line(p s)** option. The default is **linegrid(20)**, which corresponds to 20 evenly-spaced evaluation points within each bin for fitting/plotting the line.

lineplotopt(lineoption) standard graphs options to be passed on to the **twoway** command to modify the appearance of the plotted line.

Confidence Intervals

ci(ciopt) specifies the degree of polynomial and the number of smoothness constraints for constructing confidence intervals. If **ci(p s)** is specified, a piecewise polynomial of degree p with s smoothness constraints is used. If **ci(T)** is specified, **ci(1 1)** is used unless the degree p or smoothness s selection is requested via the option **pselect()** or **sselect()** (see more details in the explanation of **pselect()** and **sselect()**). If **ci(F)** or **ci()** is specified, the confidence intervals are not included in the plot. The default is **ci()**.

cigrid(cigridoption) specifies the number and location of evaluation points in the grid used to construct the confidence intervals set by the **ci(p s)** option. Two options are available: *mean* and a *numeric* non-negative integer. The option **cigrid(mean)** adds the sample average of *indvar* within each bin to the grid of evaluation points. The option **cigrid(#)** adds # number of evenly-spaced points to the grid of evaluation points for each bin. Both options can be used simultaneously: for example, **cigrid(mean 5)** generates six evaluation points within each bin containing the sample mean of *indvar* within each bin and five evenly-spaced points. The default is **cigrid(mean)**, which corresponds to one evaluation point set at the sample average of *indvar* within each bin for confidence interval construction.

ciplotopt(rcapoption) standard graphs options to be passed on to the **twoway** command to modify the appearance of the confidence intervals.

Confidence Band

cb(cbopt) specifies the degree of polynomial and the number of smoothness constraints for constructing the confidence band. If **cb(p s)** is specified, a piecewise polynomial of degree p with s smoothness constraints is used. If the option **cb(T)** is specified, **cb(1 1)** is used unless the degree p or smoothness s selection is requested via the option **pselect()** or **sselect()** (see more details in the explanation of **pselect()** and **sselect()**). If **cb(F)** or **cb()** is specified, the confidence band is not included in the plot. The default is **cb()**.

cbgrid(#) specifies the number of evaluation points of an evenly-spaced grid within each bin used for evaluation of the point estimate set by the **cb(p s)** option. The default is **cbgrid(20)**, which corresponds to 20 evenly-spaced evaluation points within each bin for confidence band construction.

cbplotopt(rareoption) standard graphs options to be passed on to the **twoway** command to modify the appearance of the confidence band.

Global Polynomial Regression

polyreg(*p*) sets the degree *p* of a global polynomial regression model for plotting. By default, this fit is not included in the plot unless explicitly specified. Recommended specification is **polyreg(3)**, which adds a cubic polynomial fit of the regression function of interest to the binned scatter plot.

polyreggrid(#) specifies the number of evaluation points of an evenly-spaced grid within each bin used for evaluation of the point estimate set by the **polyreg(*p*)** option. The default is **polyreggrid(20)**, which corresponds to 20 evenly-spaced evaluation points within each bin for confidence interval construction.

polyregcgrid(#) specifies the number of evaluation points of an evenly-spaced grid within each bin used for constructing confidence intervals based on polynomial regression set by the **polyreg(*p*)** option. The default is **polyregcgrid(0)**, which corresponds to not plotting confidence intervals for the global polynomial regression approximation.

polyregplotopt(*lineoption*) standard graphs options to be passed on to the **twoway** command to modify the appearance of the global polynomial regression fit.

Subgroup Analysis

by(*varname*) specifies the variable containing the group indicator to perform subgroup analysis; both numeric and string variables are supported. When **by(*varname*)** is specified, **binsreg** implements estimation and inference for each subgroup separately, but produces a common binned scatter plot. By default, the binning structure is selected for each subgroup separately, but see the option **samebinsby** below for imposing a common binning structure across subgroups.

bycolors(*colorstylelist*) specifies an ordered list of colors for plotting each subgroup series defined by the option **by()**.

bysymbols(*symbolstylelist*) specifies an ordered list of symbols for plotting each subgroup series defined by the option **by()**.

bylpatterns(*linepatternstylelist*) specifies an ordered list of line patterns for plotting each subgroup series defined by the option **by()**.

Binning/Degree/Smoothness Selection

nbins(*nbinsopt*) sets the number of bins for partitioning/binning of *indvar*. If **nbins(T)** or **nbins()** (default) is specified, the number of bins is selected via the companion command **binsregselect** in a data-driven, optimal way whenever possible. If a numlist with more than one number is specified, the number of bins is selected within this list via the companion command **binsregselect**.

binspos(*position*) specifies the position of binning knots. The default is **binspos(qs)**, which corresponds to quantile-spaced binning (canonical **binscatter**). Other options are: **es** for evenly-spaced binning, or a numlist for manual specification of the positions of inner knots (which must be within the range of *indvar*).

binsmethod(*method*) specifies the method for data-driven selection of the number of bins via the companion command **binsregselect**. The default is **binsmethod(dpi)**, which corresponds to the IMSE-optimal direct plug-in rule. The other option is: **rot** for rule of thumb implementation.

nbinsrot(#) specifies an initial number of bins value used to construct the DPI number of bins selector. If not specified, the data-driven ROT selector is used instead.

samebinsby forces a common partitioning/binning structure across all subgroups specified by the option **by()**. The knots positions are selected according to the option **binspos()** and using the full sample. If **nbins()** is not specified, then the number of bins is selected via the companion command **binsregselect** and using the full sample.

randcut(#) specifies the upper bound on a uniformly distributed variable used to draw a subsample for bins/degree/smoothness selection. Observations for which **runiform()** $\leq \#$ are used. $\#$ must be between 0 and 1. By default, $\max(5000, 0.01n)$ observations are used if the samples size $n > 5000$.

pselect(numlist) specifies a list of numbers within which the degree of polynomial p for point estimation is selected. Piecewise polynomials of the selected optimal degree p are used to construct dots or line if **dots(T)** or **line(T)** is specified, whereas piecewise polynomials of degree $p+1$ are used to construct confidence intervals or confidence band if **ci(T)** or **cb(T)** is specified.

sselect(numlist) specifies a list of numbers within which the number of smoothness constraints s for point estimation. Piecewise polynomials with the selected optimal s smoothness constraints are used to construct dots or line if **dots(T)** or **line(T)** is specified, whereas piecewise polynomials with $s+1$ constraints are used to construct confidence intervals or confidence band if **ci(T)** or **cb(T)** is specified. If not specified, for each value p supplied in the option **pselect()**, only the piecewise polynomial with the maximum smoothness is considered, i.e., $s=p$.

Note: To implement the degree or smoothness selection, in addition to **pselect()** or **sselect()**, **nbins(#)** must be specified.

Simulation

nsims(#) specifies the number of random draws for constructing confidence bands. The default is **nsims(500)**, which corresponds to 500 draws from a standard Gaussian random vector of size $[(p+1)J - (J-1)s]$. Setting at least **nsims(2000)** is recommended to obtain the final results.

simsgrid(#) specifies the number of evaluation points of an evenly-spaced grid within each bin used for evaluation of the supremum operation needed to construct confidence bands. The default is **simsgrid(20)**, which corresponds to 20 evenly-spaced evaluation points within each bin for approximating the supremum (or infimum) operator. Setting at least **simsgrid(50)** is recommended to obtain the final results.

simsseed(#) sets the seed for simulations.

Mass Points and Degrees of Freedom

dfcheck(n1 n2) sets cutoff values for minimum effective sample size checks, which take into account the number of unique values of *indvar* (i.e., adjusting for the number of mass points), number of clusters, and degrees of freedom of the different statistical models considered. The default is **dfcheck(20 30)**. See Cattaneo, Crump, Farrell and Feng (2025) for more details.

masspoints(masspointsoption) specifies how mass points in *indvar* are handled. By default, all mass point and degrees of freedom checks are implemented.

Available options:

masspoints(noadjust) omits mass point checks and the corresponding effective sample size adjustments.

masspoints(nolocalcheck) omits within-bin mass point and degrees of freedom checks.

masspoints(off) sets **masspoints(noadjust)** and **masspoints(nolocalcheck)** simultaneously.

masspoints(veryfew) forces the command to proceed as if *indvar* has only a few number of mass points (i.e., distinct values). In other words, forces the command to proceed as if the mass point and degrees of freedom checks were failed.

Standard Error

vce(vcetype) specifies the *vcetype* for variance estimation used by the command regress (or reghdfe if **absorb()** is specified.). The default is **vce(robust)**.

asyvar(*on/off*) specifies the method used to compute standard errors. If **asyvar**(**on**) is specified, the standard error of the nonparametric component is used and the uncertainty related to other control variables *othercovs* is omitted. Default is **asyvar**(**off**), that is, the uncertainty related to *othercovs* is taken into account.

Other Options

level(#) sets the nominal confidence level for confidence interval and confidence band estimation. Default is **level**(95).

usegtools(*on/off*) forces the use of several commands in the community-distributed Stata package **gtools** to speed the computation up, if *on* is specified. Default is **usegtools**(**off**).

For more information about the package **gtools**, please see <https://gtools.readthedocs.io/en/latest/index.html>.

precision(*single/double*) controls the storage type used for internally generated Stata variables. The default is **precision**(**double**), which uses **double** storage for temporary and generated variables to help align Stata results with the R and Python implementations. Specify **precision**(**single**) to preserve the legacy Stata implementation using **float** storage for those variables. Stata matrix and Mata computations remain double precision in either case.

noplot omits binscatter plotting, but still allows graph data to be saved with **savedata**().

savedata(*filename*) specifies a filename for saving all data underlying the binscatter plot (and more). It may be combined with **noplot**. The saved data include **bin_n**, the number of observations in each bin, and corresponding ***_n** variables for plotted series with bin identifiers.

replace overwrites the existing file when saving the graph data.

plotxrange(*min max*) specifies the range of the x-axis for plotting. Observations outside the range are dropped in the plot.

plotyrange(*min max*) specifies the range of the y-axis for plotting. Observations outside the range are dropped in the plot.

twoway options any unrecognized options are appended to the end of the twoway command generating the binned scatter plot.

Examples

Setup

```
. sysuse auto
```

Run a binscatter regression and report the plot

```
. binsreg mpg weight foreign
```

Add confidence intervals and confidence band

```
. binsreg mpg weight foreign, ci(T) cb(T) nbins(13)
```

Run binscatter regression by group

```
. binsreg mpg weight, by(foreign)
```

Stored results

Scalars

<code>e(N)</code>	number of observations
<code>e(level)</code>	confidence level
<code>e(dots_p)</code>	degree of polynomial for dots
<code>e(dots_s)</code>	smoothness of polynomial for dots
<code>e(line_p)</code>	degree of polynomial for line
<code>e(line_s)</code>	smoothness of polynomial for line
<code>e(ci_p)</code>	degree of polynomial for confidence interval
<code>e(ci_s)</code>	smoothness of polynomial for confidence interval
<code>e(cb_p)</code>	degree of polynomial for confidence band
<code>e(cb_s)</code>	smoothness of polynomial for confidence band

Matrices

<code>e(N_by)</code>	number of observations for each group
<code>e(Ndist_by)</code>	number of distinct values for each group
<code>e(Nclust_by)</code>	number of clusters for each group
<code>e(nbins_by)</code>	number of bins for each group
<code>e(cval_by)</code>	critical value for each group, used for confidence bands
<code>e(imse_var_rot)</code>	variance constant in IMSE, ROT selection
<code>e(imse_bsqr_rot)</code>	bias constant in IMSE, ROT selection
<code>e(imse_var_dpi)</code>	variance constant in IMSE, DPI selection
<code>e(imse_bsqr_dpi)</code>	bias constant in IMSE, DPI selection

References

Cattaneo, M. D., R. K. Crump, M. H. Farrell, and Y. Feng. 2024. [On Binscatter](#). *American Economic Review* 114(5): 1488-1514.

Supplemental Appendix for On Binscatter. [Supplemental Appendix](#).

Cattaneo, M. D., R. K. Crump, M. H. Farrell, and Y. Feng. 2026. [Nonlinear Binscatter Methods](#). *Review of Economics and Statistics*, revise and resubmit.

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